Exp No:11. Experiment to understand KNN algorithm for a given data set

Code:

```
import pandas as pd

data = pd.read_csv(list(uploaded.keys())[0])
data.head()

import numpy as np
import pandas as pd

data.info()
print(data['variety'].value_counts())
print(data.head())

X = data.iloc[:, 1:-1].values
y = data.iloc[:, -1].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)
```

```
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(x_train, y_train)

print(knn.score(x_train, y_train))
print(knn.score(x_test, y_test))

from sklearn.metrics import classification_report, confusion_matrix

prediction = knn.predict(x_test)
print(confusion_matrix(y_test, prediction))
print(classification_report(y_test, prediction))
```

Output:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

Result:

Thus the experiment of knn was executed successfully and the output was verified.